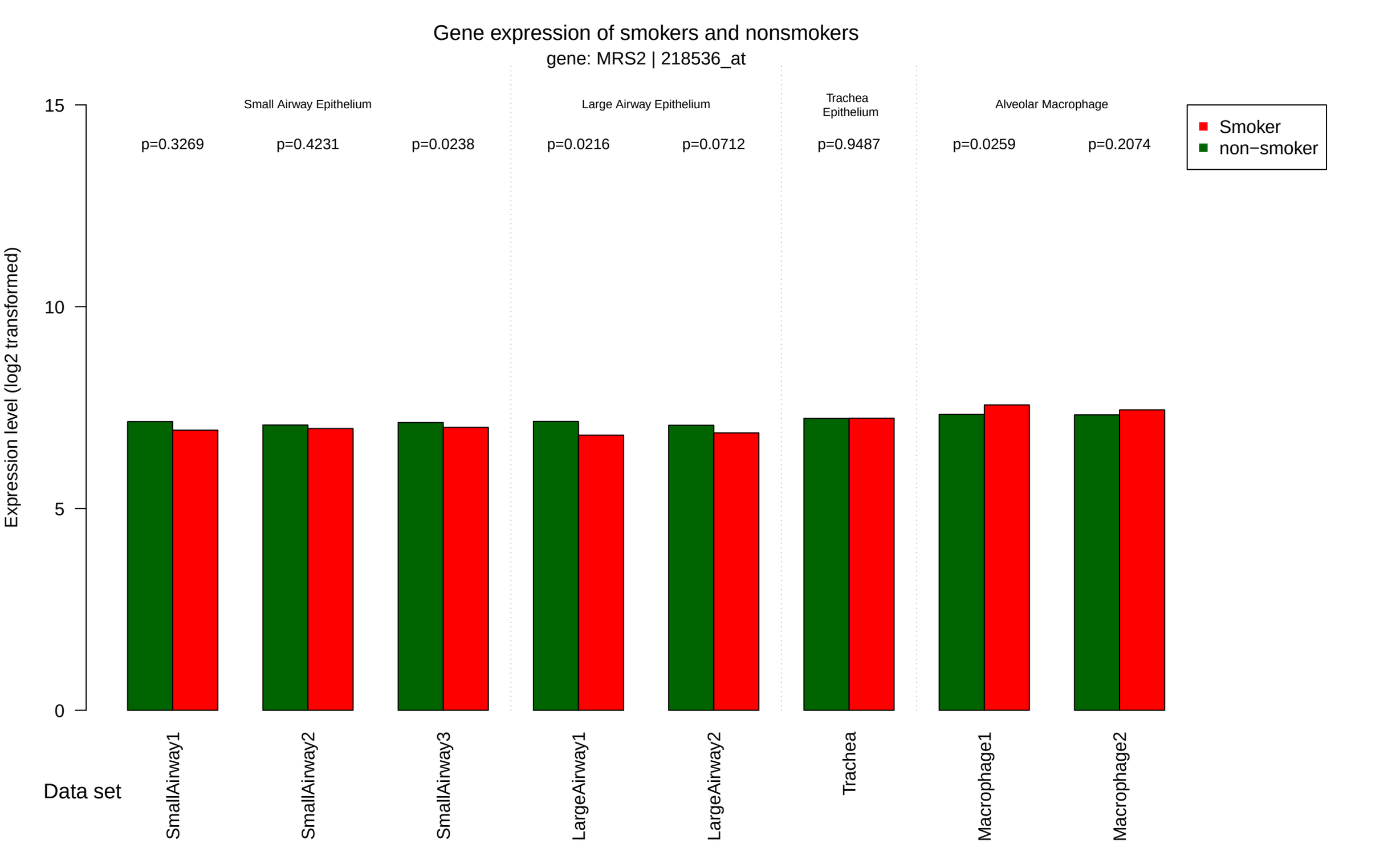
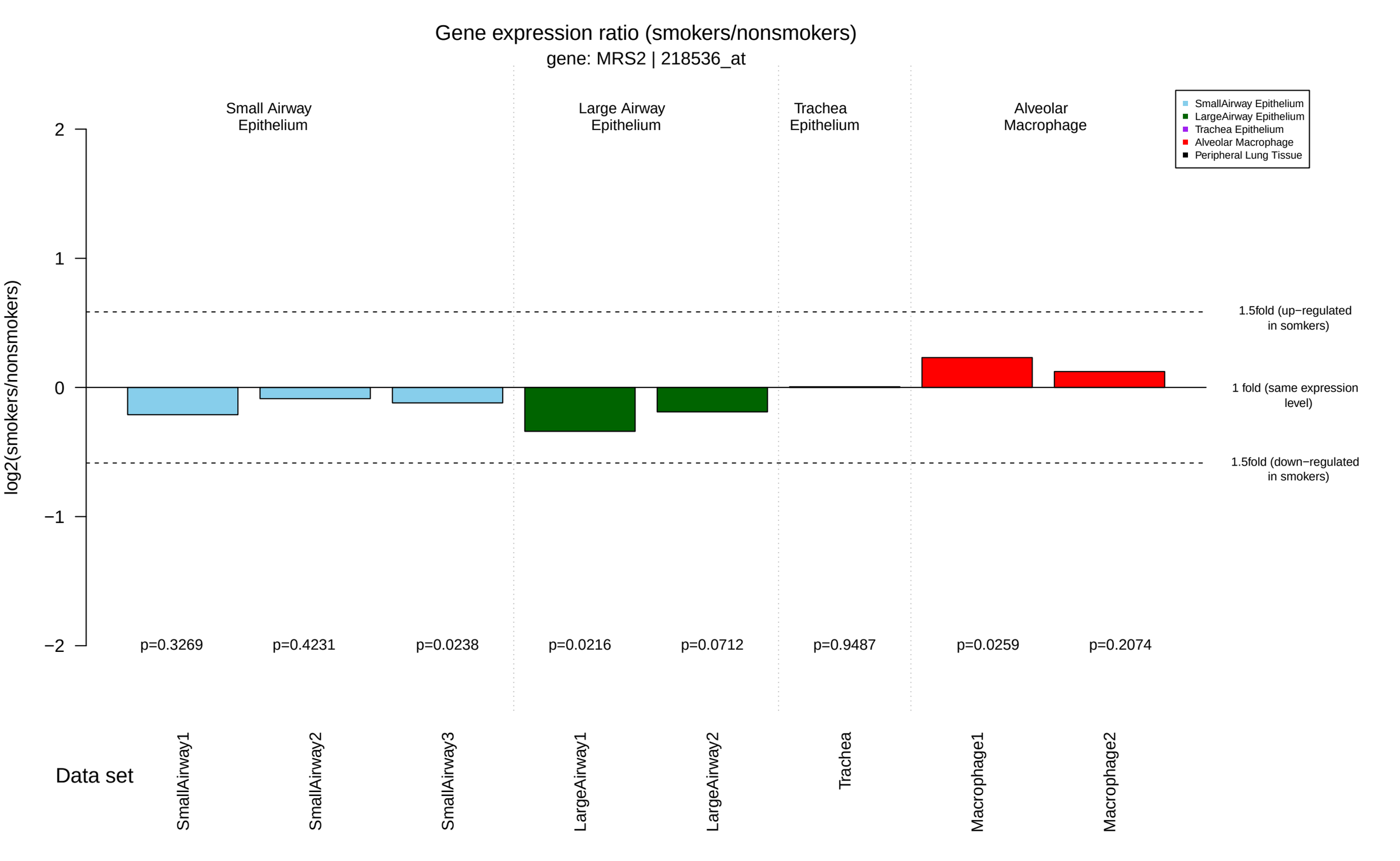
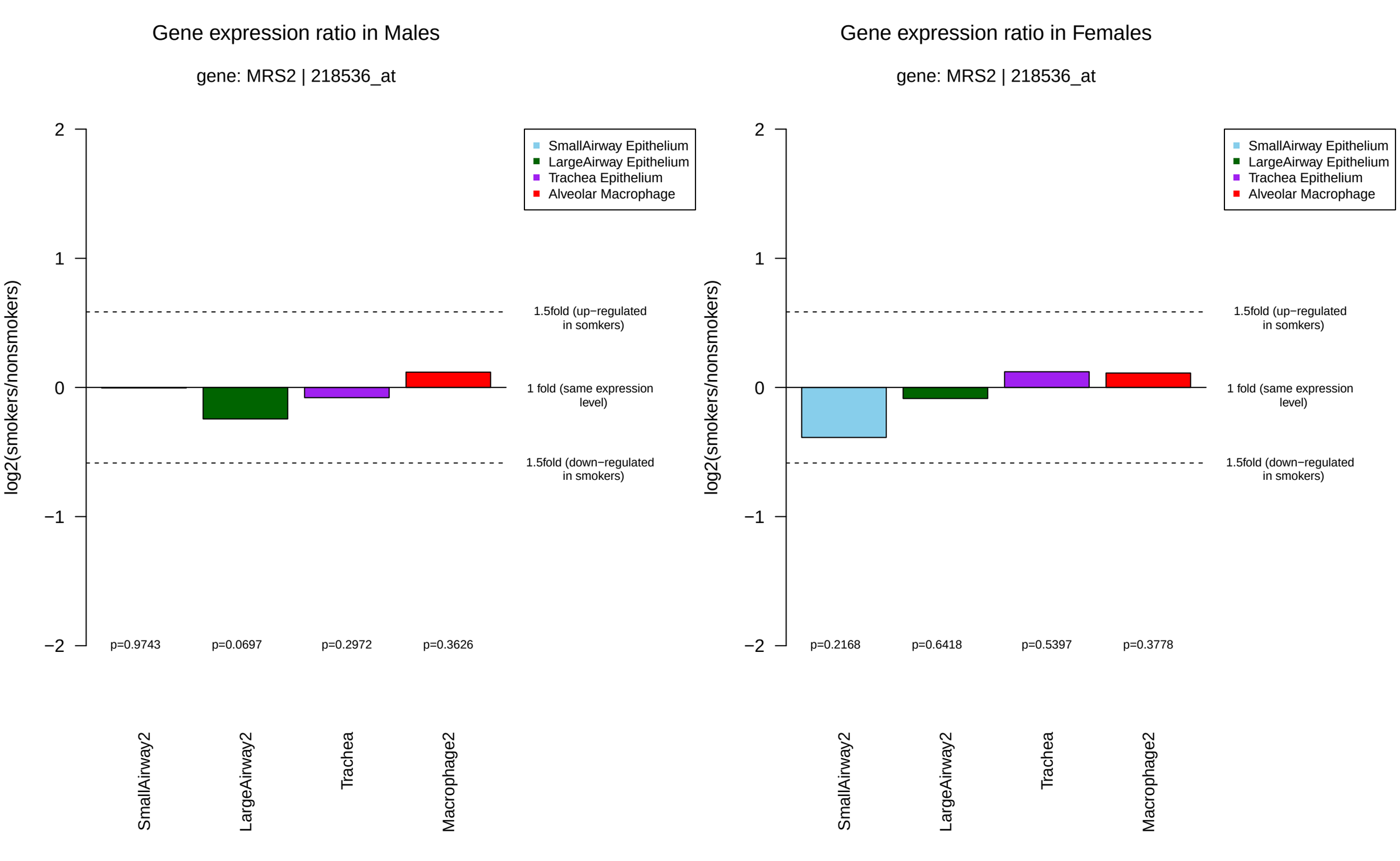
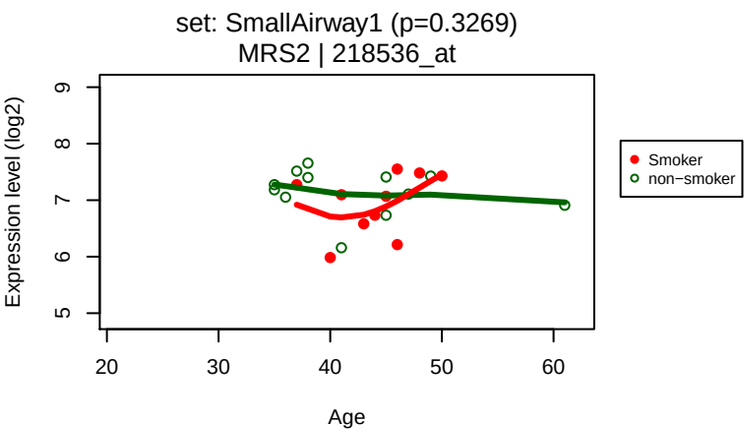
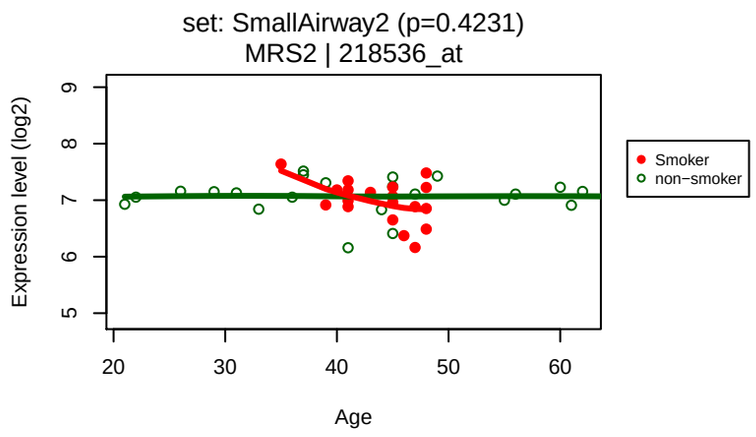
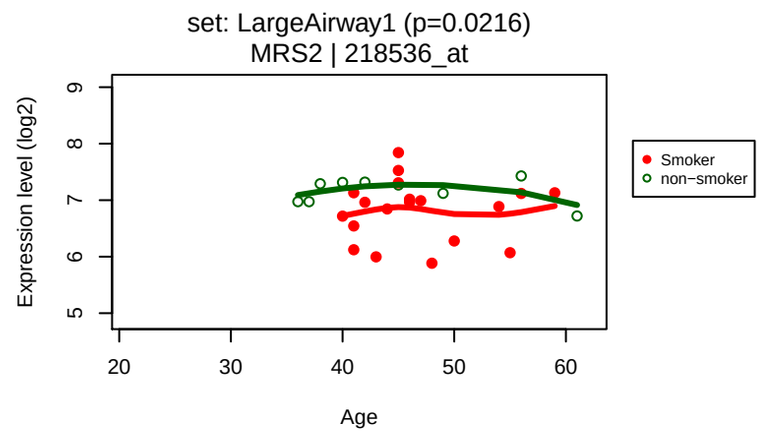

Gene expression of smokers and nonsmokers
gene: MRS2 | 218536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3269
p=0.4231
p=0.0238
p=0.0216
p=0.0712
p=0.9487
p=0.0259
p=0.2074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

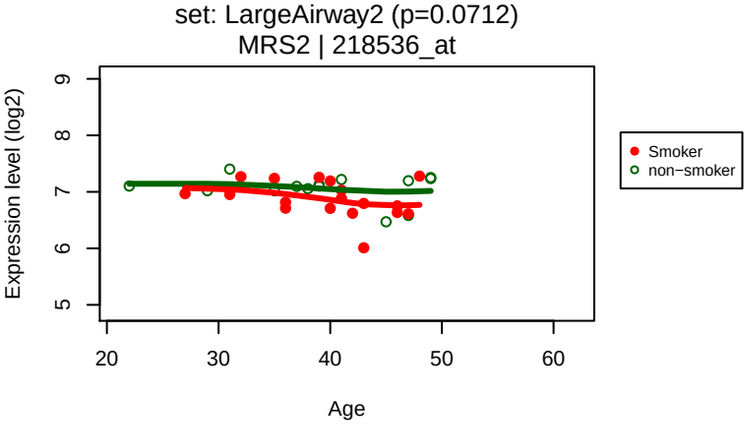

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 218536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3269
p=0.4231
p=0.0238
p=0.0216
p=0.0712
p=0.9487
p=0.0259
p=0.2074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

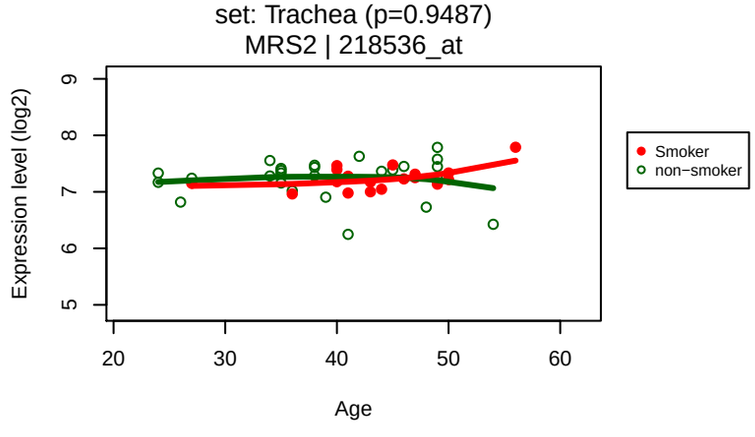

Gene expression ratio in Males
gene: MRS2 | 218536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9743
p=0.0697
p=0.2972
p=0.3626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 218536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2168
p=0.6418
p=0.5397
p=0.3778
SmallAirway2
LargeAirway2
Trachea
Macrophage2

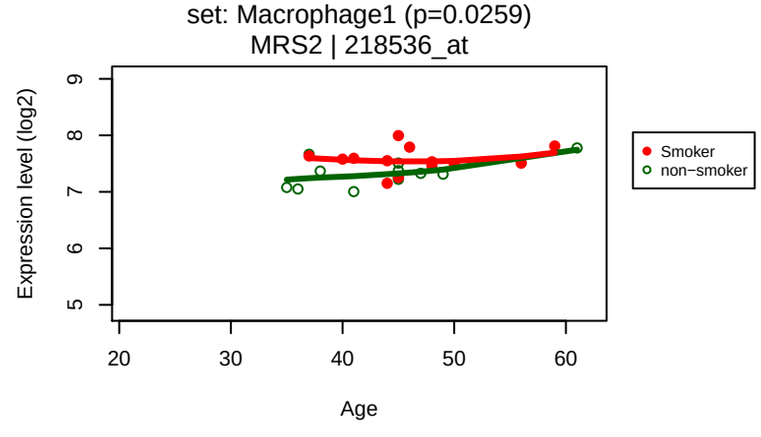

set: SmallAirway1 (p=0.3269)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker

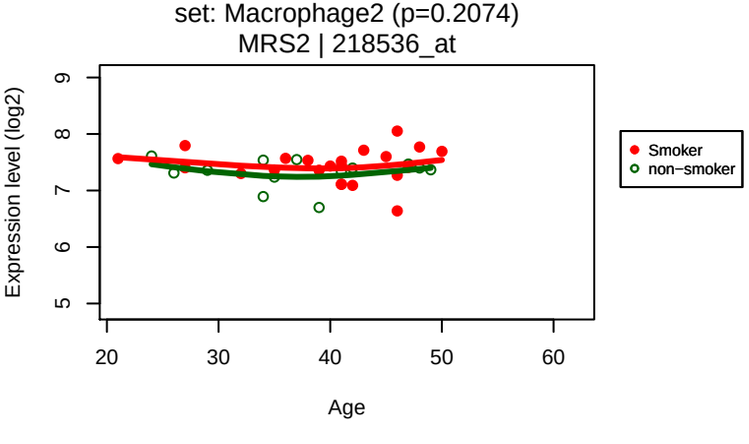

set: SmallAirway2 (p=0.4231)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker

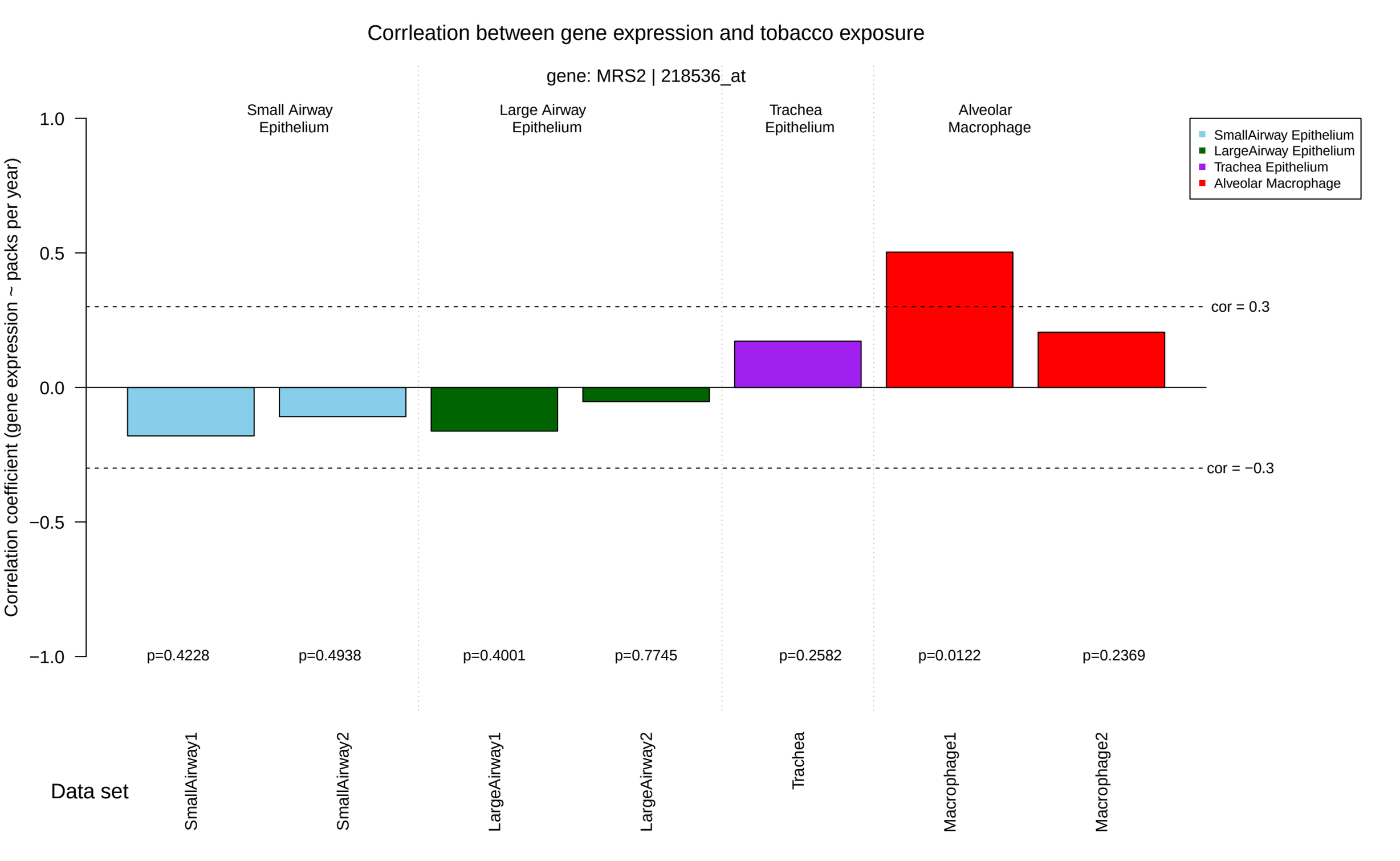

set: LargeAirway1 (p=0.0216)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0712)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9487)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0259)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2074)
MRS2 | 218536_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 218536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4228
p=0.4938
p=0.4001
p=0.7745
p=0.2582
p=0.0122
p=0.2369
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2